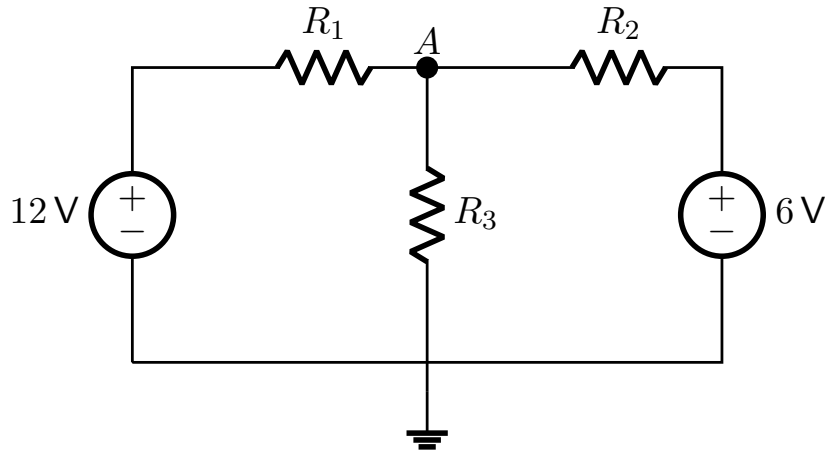


Example 1 — Two-source circuit

Show your work and include units. All sources and wires are ideal. All node voltages are measured relative to ground. Solutions to Examples 1–7 begin after Example 7.

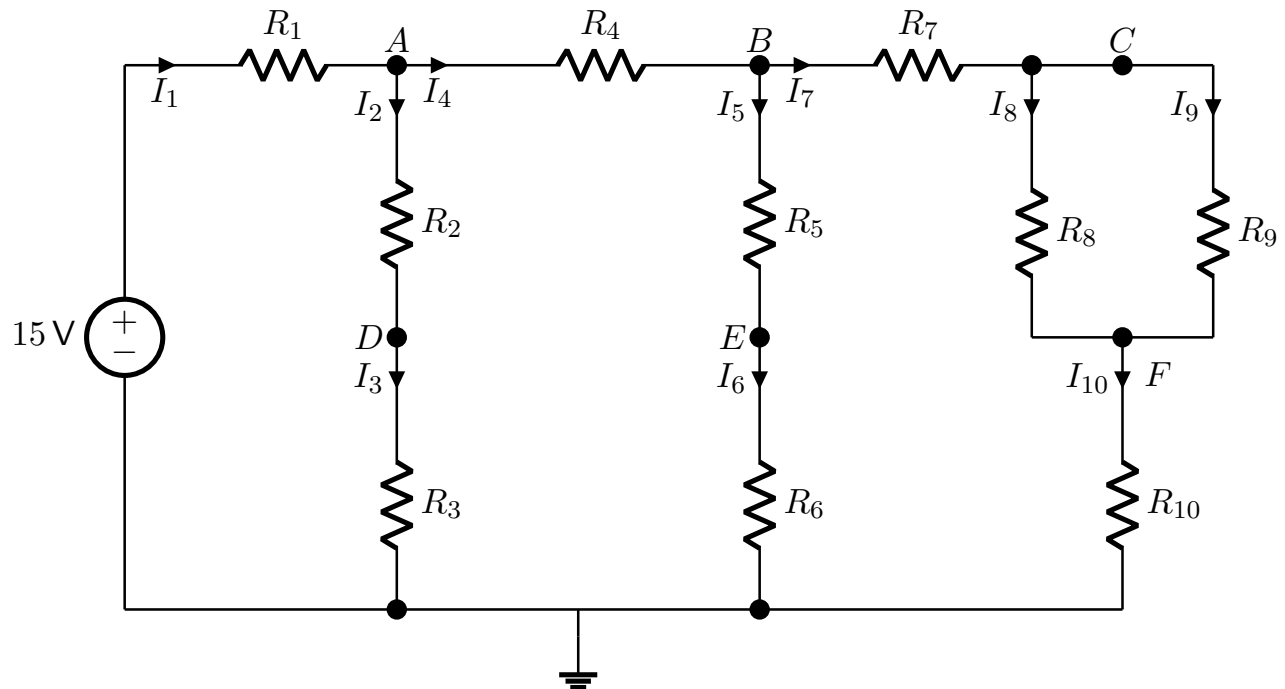
All three resistors have resistance $2\text{ k}\Omega$. Define I_1 from the 12 V source toward A, I_2 from A toward the 6 V source, and I_3 from A toward ground.



- Express I_1 , I_2 , and I_3 in terms of V_A . Write a KCL equation at A.
- Solve for V_A and all three currents. Explain the value obtained for I_2 .
- Write KVL around each of the two inner loops and verify both equations numerically.

Example 2 — Series and parallel reduction

All 10 resistors are $1\text{ k}\Omega$. Use successive series and parallel reductions. Label any intermediate equivalent resistances you use.



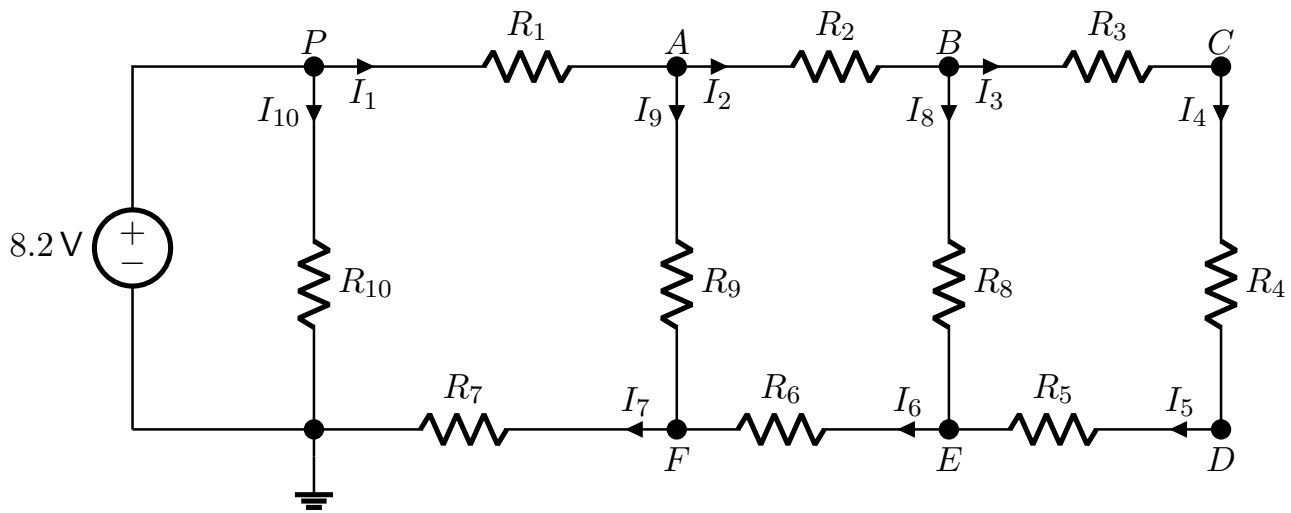
- Find the equivalent resistance seen by the voltage source. Show the reduction steps.
- Find the total current supplied by the voltage source.
- Find the voltage at each of the six labeled nodes. Show your calculations.

Node	A	B	C	D	E	F
Voltage (V)						

Use additional paper for calculations as needed.

Example 3 — KCL, KVL, and branch currents

All 10 resistors are $1\text{ k}\Omega$. Use the indicated current directions. I_s is the total current supplied by the voltage source.



- Write KCL at each node A through F, and relate I_s to the currents leaving node P.
- Choose four independent loops. Identify each loop by its elements or node sequence, and write one KVL equation for each.
- Calculate I_1 through I_{10} and I_s . Show your intermediate calculations.
- Substitute your numerical currents into one KCL equation at a branching node and one KVL equation. Verify that both are satisfied.

Current	I_1	I_2	I_3	I_4	I_5
mA					

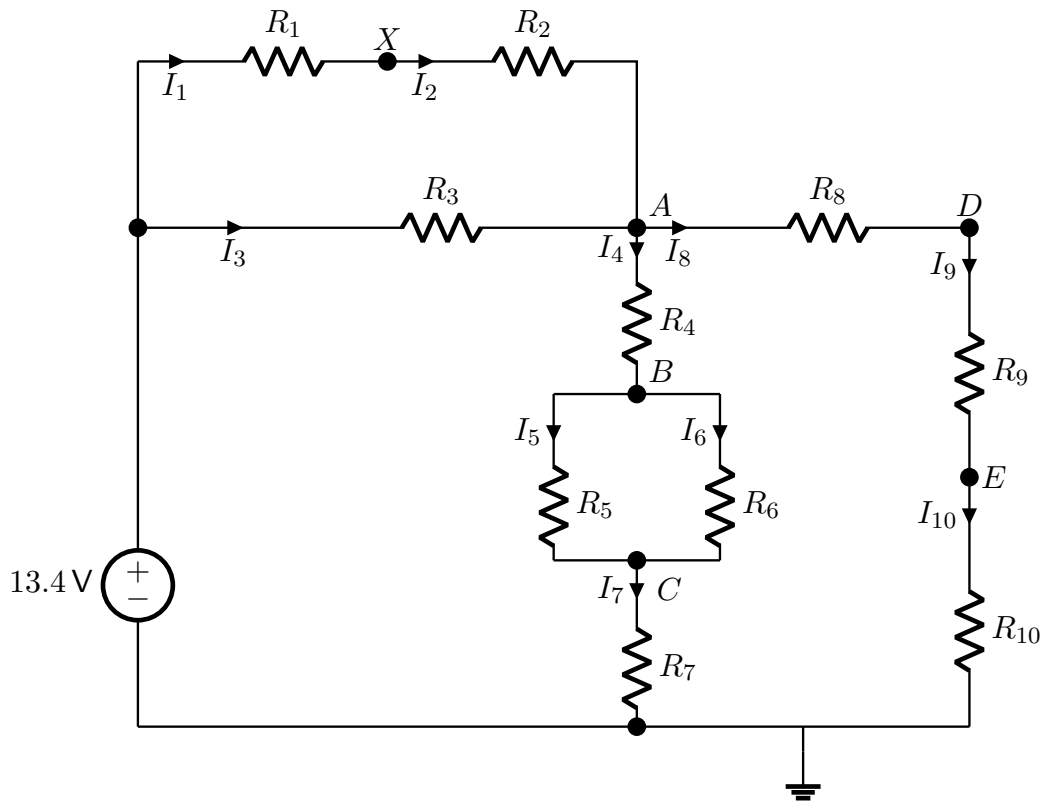
Current	I_6	I_7	I_8	I_9	I_{10}
mA					

$$I_s = \text{_____ mA}$$

Use additional paper for calculations as needed.

Example 4 — Series and parallel reduction

All 10 resistors are $1\text{ k}\Omega$. Use successive series and parallel reductions. Label any intermediate equivalent resistances you use.



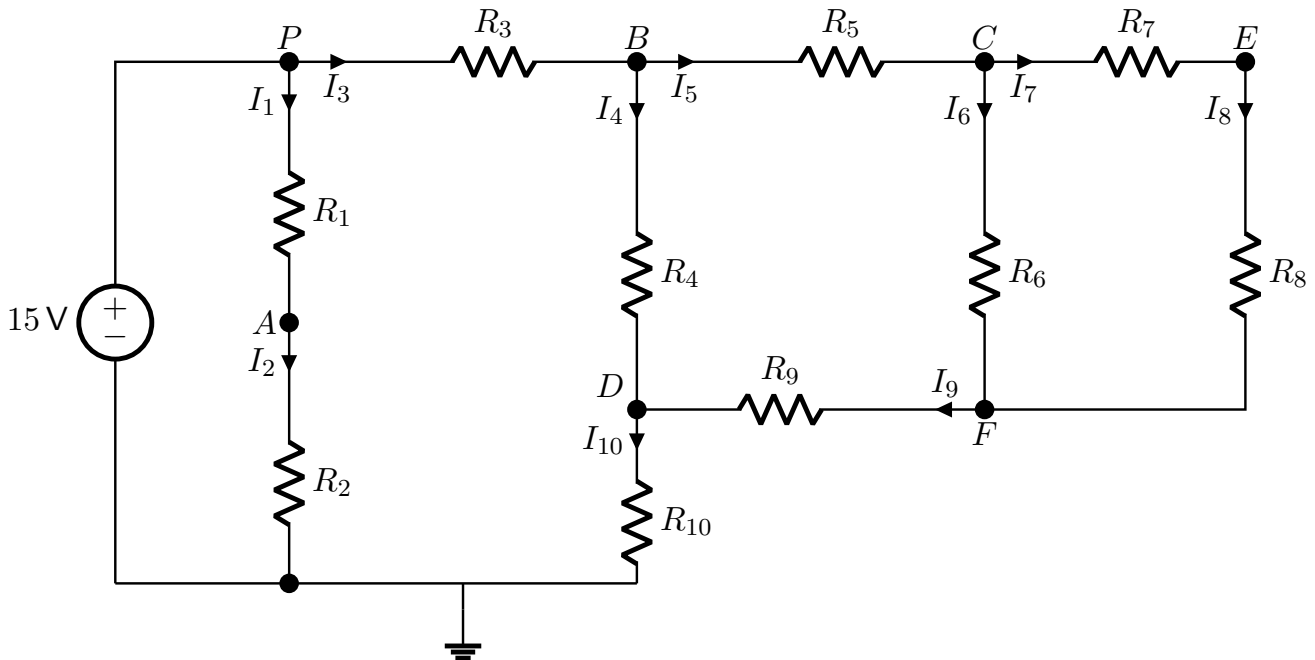
- Find the equivalent resistance seen by the voltage source. Show the reduction steps.
- Find the total current supplied by the voltage source.
- Find the voltage at each of the six labeled nodes. Show your calculations.

Node	A	B	C	D	E	X
Voltage (V)						

Use additional paper for calculations as needed.

Example 5 — KCL, KVL, and branch currents

All 10 resistors are $1\text{ k}\Omega$. Use the indicated current directions. I_s is the total current supplied by the voltage source.



- Write KCL at each node A through F, and relate I_s to the currents leaving node P.
- Choose four independent loops. Identify each loop by its elements or node sequence, and write one KVL equation for each.
- Calculate I_1 through I_{10} and I_s . Show your intermediate calculations.
- Substitute your numerical currents into one KCL equation at a branching node and one KVL equation. Verify that both are satisfied.

Current	I_1	I_2	I_3	I_4	I_5
mA					

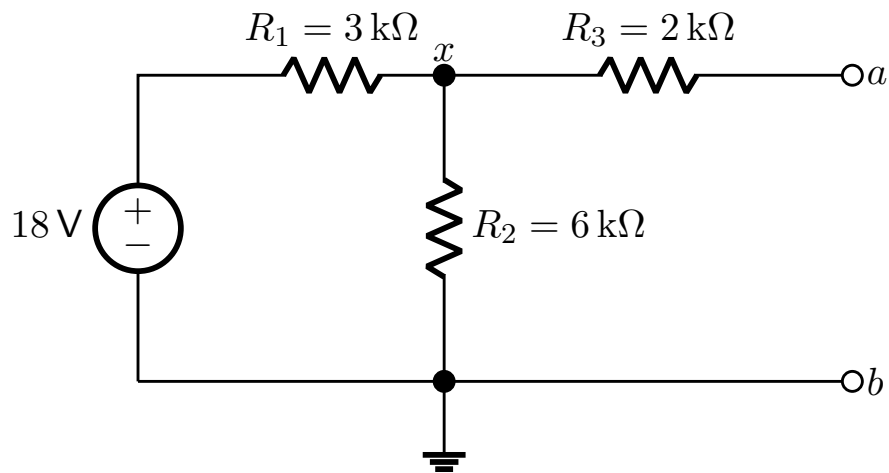
Current	I_6	I_7	I_8	I_9	I_{10}
mA					

$$I_s = \text{_____ mA}$$

Use additional paper for calculations as needed.

Example 6 — Thévenin equivalent and load current

All sources and wires are ideal. Terminal b is ground. The network is initially unloaded at terminals a–b. Use the resistor values shown.

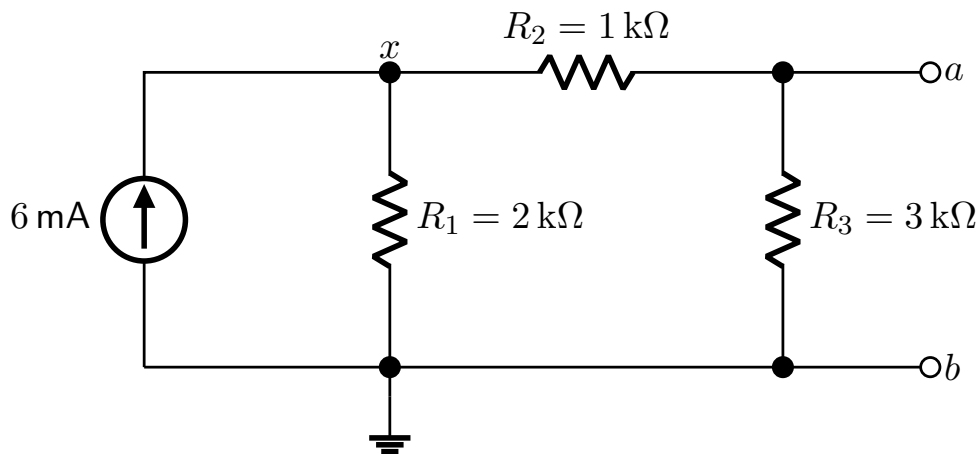


- With a–b open, find the open-circuit voltage V_{ab} . Explain why no current flows through R_3 .
- Deactivate the independent source and find the resistance seen looking into a–b. Draw the Thévenin equivalent, labeling its voltage polarity.
- Find and draw the Norton equivalent, including the current-source direction. Verify its current by directly shorting a–b in the original circuit.
- Connect an $8\text{ k}\Omega$ load between a and b. Use an equivalent circuit to calculate the load current from a to b and the voltage across the load.

Show your calculations. Use additional paper as needed.

Example 7 — Norton equivalent and source conversion

All sources and wires are ideal. Terminal b is ground. The network is initially unloaded at terminals a–b. Use the resistor values shown.



- Short terminals a and b. Find the short-circuit current from a to b, showing how the source current divides.
- Remove the short, deactivate the independent source, and find the resistance seen looking into a–b. Draw the Norton equivalent, labeling its current-source direction.
- Convert your Norton equivalent to a Thévenin equivalent. Verify its voltage by finding V_{ab} directly in the original unloaded circuit.
- Connect a $4.5\text{ k}\Omega$ load between a and b. Use the Norton equivalent to find the load current from a to b and the load voltage.

Show your calculations. Use additional paper as needed.

Solution 1 — Two-source circuit

a) Current expressions and KCL

$$I_1 = \frac{12\text{ V} - V_A}{2\text{ k}\Omega}, \quad I_2 = \frac{V_A - 6\text{ V}}{2\text{ k}\Omega}, \quad I_3 = \frac{V_A}{2\text{ k}\Omega}$$

$$I_1 = I_2 + I_3$$

b) Node voltage and currents

Substitute the current expressions into KCL and multiply by $2\text{ k}\Omega$:

$$12\text{ V} - V_A = V_A - 6\text{ V} + V_A \Rightarrow V_A = 6\text{ V}$$

$$I_1 = 3\text{ mA}, \quad I_2 = 0, \quad I_3 = 3\text{ mA}$$

Both terminals of R_2 are at 6 V , so its voltage drop and current are zero. R_2 is still connected; it is not an open circuit.

c) KVL and numerical checks

Left loop: rise through the 12 V source, then drops through R_1 and R_3 .

$$12\text{ V} - I_1 R_1 - I_3 R_3 = 0 \Rightarrow 12 - 6 - 6 = 0\text{ V}$$

Right loop: traverse R_3 from ground to A, then R_2 toward the 6 V source, and return through that source to ground.

$$I_3 R_3 - I_2 R_2 - 6\text{ V} = 0 \Rightarrow 6 - 0 - 6 = 0\text{ V}$$

Solution 2 — Series and parallel reduction

The parallel symbol below denotes an equivalent parallel combination. All intermediate resistances are in $\text{k}\Omega$.

$$R_{C0} = (R_8 \parallel R_9) + R_{10} = 1.5$$

$$R_{B0} = (R_5 + R_6) \parallel (R_7 + R_{C0}) = 2 \parallel 2.5 = \frac{10}{9}$$

$$R_{A0} = (R_2 + R_3) \parallel (R_4 + R_{B0}) = 2 \parallel \frac{19}{9} = \frac{38}{37}$$

$$R_{eq} = R_1 + R_{A0} = \frac{75}{37}\text{ k}\Omega \approx 2.027\text{ k}\Omega$$

$$I_s = \frac{15}{75/37} = 7.4\text{ mA}, \quad V_A = 15 - 7.4 = 7.6\text{ V}$$

The A-to-B branch carries $7.6/(19/9) = 3.6\text{ mA}$, so $V_B = 7.6 - 3.6 = 4\text{ V}$. The B-to-C branch carries $4/2.5 = 1.6\text{ mA}$, so $V_C = 2.4\text{ V}$. The A-to-ground and B-to-ground series pairs divide their respective voltages equally, giving $V_D = 3.8\text{ V}$ and $V_E = 2\text{ V}$. R_{10} carries 1.6 mA , giving $V_F = 1.6\text{ V}$.

Node	A	B	C	D	E	F
Voltage (V)	7.6	4	2.4	3.8	2	1.6

Solution 3 — KCL, KVL, and branch currents

KCL equations

$$A : I_1 = I_2 + I_9 \quad ; \quad B : I_2 = I_3 + I_8$$

$$C : I_3 = I_4 \quad ; \quad D : I_4 = I_5$$

$$E : I_5 + I_8 = I_6 \quad ; \quad F : I_6 + I_9 = I_7$$

$$P : I_s = I_1 + I_{10}$$

One valid set of four independent KVL equations

$$8.2 - I_{10}R_{10} = 0$$

$$I_{10}R_{10} - I_1R_1 - I_9R_9 - I_7R_7 = 0$$

$$I_9R_9 - I_2R_2 - I_8R_8 - I_6R_6 = 0$$

$$I_8R_8 - I_3R_3 - I_4R_4 - I_5R_5 = 0$$

These are the four adjacent meshes, from left to right. With resistance in $k\Omega$, the inner reductions are $1 \parallel 3 = 3/4$, then $1 \parallel (1 + 3/4 + 1) = 11/15$. The source sees $1 \parallel (1 + 11/15 + 1) = 41/56 k\Omega$, so $I_s = 11.2 \text{ mA}$. R_{10} carries 8.2 mA and the R_1 branch carries 3 mA . The A-to-F voltage is $3 \times (11/15) = 2.2 \text{ V}$, giving $I_9 = 2.2 \text{ mA}$. Thus $I_2 = I_6 = 0.8 \text{ mA}$. The B-to-E voltage is $0.8 \times (3/4) = 0.6 \text{ V}$, so $I_8 = 0.6 \text{ mA}$ and $I_3 = I_4 = I_5 = 0.2 \text{ mA}$. Finally, $I_7 = I_1 = 3 \text{ mA}$.

Current	I_1	I_2	I_3	I_4	I_5
mA	3	0.8	0.2	0.2	0.2

Current	I_6	I_7	I_8	I_9	I_{10}
mA	0.8	3	0.6	2.2	8.2

Numerical checks: at A, $3 = 0.8 + 2.2 \text{ mA}$. In the rightmost mesh, $0.6 - 0.2 - 0.2 - 0.2 = 0 \text{ V}$. All currents are positive in the indicated directions.

Solution 4 — Series and parallel reduction

Let P denote the positive terminal of the source. Node 0 denotes ground.

The parallel symbol below denotes an equivalent parallel combination. All intermediate resistances are in $\text{k}\Omega$.

$$R_{PA} = (R_1 + R_2) \parallel R_3 = 2 \parallel 1 = \frac{2}{3}$$

$$R_{BC} = R_5 \parallel R_6 = \frac{1}{2}$$

$$R_{A0} = (R_4 + R_{BC} + R_7) \parallel (R_8 + R_9 + R_{10}) = 2.5 \parallel 3 = \frac{15}{11}$$

$$R_{eq} = R_{PA} + R_{A0} = \frac{67}{33} \text{ k}\Omega \approx 2.030 \text{ k}\Omega$$

$$I_s = \frac{13.4}{67/33} = 6.6 \text{ mA}, \quad V_A = 6.6 \left(\frac{15}{11} \right) = 9 \text{ V}$$

The P-to-A voltage drop is 4.4 V. The upper two-resistor path carries 2.2 mA, giving $V_X = 11.2 \text{ V}$. From A to ground, the $2.5 \text{ k}\Omega$ branch carries 3.6 mA and the $3 \text{ k}\Omega$ branch carries 3 mA. Therefore $V_B = 9 - 3.6 = 5.4 \text{ V}$, $V_C = 3.6 \text{ V}$, $V_D = 9 - 3 = 6 \text{ V}$, and $V_E = 3 \text{ V}$.

Node	A	B	C	D	E	X
Voltage (V)	9	5.4	3.6	6	3	11.2

Solution 5 — KCL, KVL, and branch currents

KCL equations

$$\begin{aligned}
 A : I_1 &= I_2 & ; & & B : I_3 &= I_4 + I_5 \\
 C : I_5 &= I_6 + I_7 & ; & & D : I_4 + I_9 &= I_{10} \\
 E : I_7 &= I_8 & ; & & F : I_6 + I_8 &= I_9 \\
 P : I_s &= I_1 + I_3
 \end{aligned}$$

One valid set of four independent KVL equations

$$\begin{aligned}
 15 - I_1 R_1 - I_2 R_2 &= 0 \\
 15 - I_3 R_3 - I_4 R_4 - I_{10} R_{10} &= 0 \\
 I_4 R_4 - I_5 R_5 - I_6 R_6 - I_9 R_9 &= 0 \\
 I_6 R_6 - I_7 R_7 - I_8 R_8 &= 0
 \end{aligned}$$

The first two loops contain the source and the indicated resistor paths. The last two are the B-C-F-D and C-E-F loops. With resistance in $\text{k}\Omega$, $R_{CF} = R_6 \parallel (R_7 + R_8) = 2/3$. Then $R_{BD} = R_4 \parallel (R_5 + R_{CF} + R_9) = 8/11$. The source sees $2 \parallel (1 + 8/11 + 1) = 15/13 \text{ k}\Omega$, giving $I_s = 13 \text{ mA}$. The R_3 branch carries 5.5 mA , giving $V_B - V_D = 4 \text{ V}$. Thus $I_4 = 4 \text{ mA}$ and $I_5 = 1.5 \text{ mA}$; the last split gives $I_6 = 1 \text{ mA}$ and $I_7 = 0.5 \text{ mA}$.

Current	I_1	I_2	I_3	I_4	I_5
mA	7.5	7.5	5.5	4	1.5

Current	I_6	I_7	I_8	I_9	I_{10}
mA	1	0.5	0.5	1.5	5.5

Numerical checks: at B, $5.5 = 4 + 1.5 \text{ mA}$. In the C-E-F loop, $1 - 0.5 - 0.5 = 0 \text{ V}$. All currents are positive in the indicated directions.

Solution 6 — Thévenin and Norton equivalents

a) Open-circuit voltage

No load is connected, so R_3 carries zero current and has zero voltage drop. Thus the voltage at a equals the divider voltage at x.

$$V_{Th} = V_{ab} = 18 \frac{6}{3+6} = 12 \text{ V}$$

b) Equivalent resistance

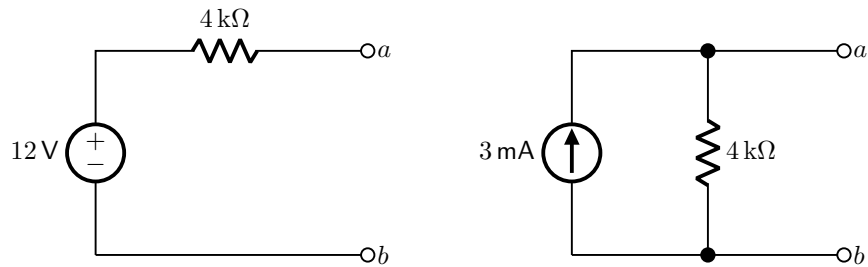
Replace the ideal voltage source with a short circuit. R_1 and R_2 are then in parallel, in series with R_3 .

$$R_{Th} = R_3 + (R_1 \parallel R_2) = 2 + \frac{3 \times 6}{3+6} = 4 \text{ k}\Omega$$

c) Norton current and short-circuit check

$$I_N = \frac{V_{Th}}{R_{Th}} = \frac{12}{4} = 3 \text{ mA}, \quad R_N = R_{Th} = 4 \text{ k}\Omega$$

With a–b shorted in the original circuit, R_2 and R_3 are parallel: $6 \parallel 2 = 1.5 \text{ k}\Omega$. The source current is $18/(3+1.5) = 4 \text{ mA}$. Current division gives $4 \times 6/(6+2) = 3 \text{ mA}$ through R_3 and the external short, from a to b. The Norton source itself points from b toward a.



Thévenin equivalent (left) and Norton equivalent (right).

d) Load connected

$$I_L = \frac{12}{4+8} = 1 \text{ mA}, \quad V_{ab} = I_L R_L = 8 \text{ V}$$

Solution 7 — Norton and Thévenin equivalents

a) Short-circuit current

Shorting a–b bypasses R_3 . The 6 mA source divides between $R_1 = 2\text{ k}\Omega$ and $R_2 = 1\text{ k}\Omega$. The current through R_2 flows through the external short from a to b.

$$I_N = I_{sc} = 6 \frac{2}{2+1} = 4\text{ mA}$$

b) Equivalent resistance

Replace the ideal current source with an open circuit. The R_2 – R_1 series path is in parallel with R_3 .

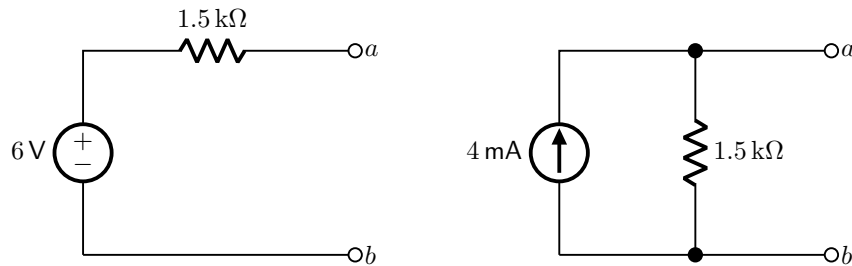
$$R_N = (R_2 + R_1) \parallel R_3 = (1 + 2) \parallel 3 = 1.5\text{ k}\Omega$$

The Norton source points from b toward a, in parallel with $1.5\text{ k}\Omega$.

c) Thévenin conversion and open-circuit check

$$V_{Th} = I_N R_N = 4 \times 1.5 = 6\text{ V}, \quad R_{Th} = 1.5\text{ k}\Omega$$

In the original unloaded circuit, R_1 is in parallel with the R_2 – R_3 series path. The current in that series path is $6 \times 2/(2 + 1 + 3) = 2\text{ mA}$. Hence $V_{ab} = (2\text{ mA})(3\text{ k}\Omega) = 6\text{ V}$, positive at a.



Thévenin equivalent (left) and Norton equivalent (right).

d) Load connected: Norton current division

$$I_L = I_N \frac{R_N}{R_N + R_L} = 4 \frac{1.5}{1.5 + 4.5} = 1\text{ mA}$$

$$V_{ab} = I_L R_L = 4.5\text{ V}$$